

essential idea in the dynamics of strings in background fields. That is that the space-time equations of motion for the background field are given by the requirement that the Wilson line boundary term in the BCFT preserves the conformal invariance of the world sheet theory. To determine the spacetime equations of motion for the gauge field, we compute the CFT beta function associated with the boundary term and set it equal to zero. The appropriate beta function is defined by introducing a renormalization counterterm for the Wilson line boundary interaction of the form

$$\Delta S = \frac{i}{2\pi} \int_{\sigma=0} d\tau \Gamma_\mu \partial_\tau X^\mu \quad (38)$$

In the CFT calculation of the counterterm, an ultraviolet cutoff Λ must be introduced. The beta function which determines the conformal symmetry of the boundary term is then

$$\beta_\mu = \Lambda \frac{\partial}{\partial \Lambda} \Gamma_\mu \quad (39)$$

The requirement that this beta function should vanish leads to the spacetime gauge field EOM,

$$(\nabla^\nu \hat{F}_\mu{}^\lambda)(1 - \hat{F}^2)_{\lambda\nu}^{-1} = 0 \quad (40)$$

where $\hat{F} \equiv 2\pi\alpha' F$ and $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. To leading order in α' , this is just Maxwell's equation. In fact, the exact equation of motion (40) can be derived from an effective gauge action of the Dirac-Born-Infeld (DBI) form,

$$\mathcal{L}_{eff} \propto \sqrt{\det(1 + \hat{F})} \quad (41)$$

In the $\alpha' \rightarrow 0$ limit, this coincides with the large N result $\mathcal{L}_{eff} \propto Tr F^2$. But the fact that the full action in the string-theoretic formulation is of the DBI form has very interesting implications for the dynamics of electric flux-string formation [20]. Since the Coulomb force in two dimensions is linearly confining even in perturbation theory, one might not expect the confinement mechanism exhibited by the CP^{N-1} model to be particularly illuminating for 4-dimensional QCD. But the connection between 2- and 4-dimensional confinement appears to be much closer if we view confinement in the CP^{N-1} model in the context of brane dynamics in a tachyonic crystal vacuum. Electric flux tubes are associated with transverse motion of the branes (since they are fundamental strings). Thus the spatial dimensionality of the electric flux distribution between charges is determined by the codimensionality of the branes, suggesting a straightforward generalization of this mechanism to 4-dimensional QCD. As discussed in Ref. [20] linear strings of electric flux arise from the DBI action by considering a point-charge source for a gauge field living on a D-brane. In order to construct a BPS configuration, the source must be accompanied by a transverse fluctuation of the